

ERASTEEL TECHNICAL DATASHEET: Evoloop M42

Application Family: Cutting tools | Standards: AISI M42, W.-Nr. 1.3247, HS 2-10-1-8

1. Grade Overview & Characteristics

Evoloop M42 is Erasteel's conventional cobalt high-speed steel with 8% cobalt and 9.4% molybdenum. Exhibits outstanding red hardness and high working hardness for cost-effective heavy machining.

2. Nominal Chemical Composition (wt. % range)

Element	Carbon (C)	Chromium (Cr)	Molybdenum (Mo)	Tungsten (W)	Vanadium (V)	Cobalt (Co)	Other
Range %	1.05 - 1.11	3.6 - 4.0	9.2 - 9.6	1.3 - 1.7	1.1 - 1.3	7.8 - 8.2	Conventional ingot-cast / ESR HSS

3. Mechanical Hardness & Property Profiles (Tempered state)

Metric	Value	Metric	Value (Rating 1-10)
Working Hardness Range	66.0 - 68.0 HRC	Machinability	6.5/10
Max Reachable Hardness	68.0 HRC	Wear Resistance	7.0/10
Annealed Delivery State	Max 280-350 HB	Toughness	5.0/10
Hot Hardness Retention	Secondary hardening	Hot Hardness	8.5/10
Heat Treatment Type	Vacuum / Salt bath	Grindability	6.0/10

4. Industrial Applications

Bi-metal bandsaw blade teeth, twist drills for aerospace titanium alloys, broaches, milling cutters, reamers.

5. Heat Treatment & Tempering Guidelines

Austenitizing at 1170-1200°C. Triple tempering at 540-560°C. Reachable hardness 68 HRC.